Types of Numbers

1) Composite Numbers : Any Number that has more than 2 factor is a Composite number.

{4, 6, 8, 10, ...}

2) Natural Numbers: Any Number that one used as Counting numbers are natural numbers
{1, 2, 3, 4, ...}

3) ~~Perfect Number~~ Integers: We Use negative numbers to show number len than zero ("0") Combining these whole numbers, We Obtain a new Collection of numbers {-1, 0, 1, 2, 3} not Include fraction(on) decimal.

4) Whole Numbers: To Describe, The Counting number (on) natural numbers along With Zero form Whole number {0, 1, 2, 3}

5) Real : Real numbers Include all rational numbers and Irrational numbers. The numbers that Can be represented on a real numbe line represent a real number.

Sets

- **Sets?**
- A Sets is a well defined Collection of distinct objects.

Ex:-

⇒ The Collection of first five prime natural number is a Set Containing the element 2, 3, 5, 7, 11

→ To understand

There Should be a set of object which should be well defined. {like they belong to something that's when we can call them sets}

Representing Sets

1) Roster Form: In this method a Set is described by listing elements, separated Commas & then Enclose them by Curly bracket

B ⇒ {1, 2, 3, 4, 5}

Capital Letter

Note: While Writing the Set in roster form, an Element is not generally repeated.

Ex:

The Set of letters of Word MISSISSIPI may be written as

$A \Rightarrow \{M, I, S, P\}$

not Repeated.

2) SET Builder format:

In this We Write down a property (or) rule which gives us all the element of the Set.

$\therefore A \Rightarrow \{x : P(x)\}$

Where $P(x)$ in the property by which $x \in A$ and Colon (:) Stands for 'Such that'

Ex) Set $A \Rightarrow \{x^3 : x < 5,\ x \in \text{Whole numbers}\}$ in roster form.

→ 1) $A \Rightarrow \{x^3 : x < 5,\ x \in \text{Whole numbers}\}$
Given,
$x < 5,\ x \in \text{Whole numbers } \{W\}$

2) $x \geq 0$ {Cause of Whole numbers}
$x < 5$ {Given}

$x \Rightarrow 0, 1, 2, 3, 4$

3) Roster form ⇒
$A \Rightarrow \{0, 1, 2, 3, 4\}$ { this is the value of x }

$x^3 \Rightarrow ?$

ans. $A \Rightarrow \{0, 1, 8, 27, 64\}$

Ex) Express Set $B \Rightarrow \{2, 5, 10, 17, 26\}$ in Set builder form.
Given,
→ $B \Rightarrow \{2, 5, 10, 17, 26\}$
~~Given~~.

1) Set Builder form
$\{x : \text{Property of } x\}$

2) $B \Rightarrow \{2, 5, 10, 17, 26\}$
$x = n^2 + 1,\ n \in N,\ n \leq 5$

3) $B \Rightarrow \{x : x = n^2 + 1,\ n \in N$
$n \leq 5$

Ex. Express Set $C \Rightarrow \left\{\frac{1}{9}, \frac{1}{25}, \frac{1}{49}, \frac{1}{81}\right\}$ in Set builder form.

→ Given

1) $C \Rightarrow \left\{\frac{1}{9}, \frac{1}{25}, \frac{1}{49}, \frac{1}{81}\right\}$

2) Set builder form

$\frac{1}{9} \Rightarrow \frac{1}{(2n+1)^2}$

$n \in N$

$n \leq 4$

$\frac{1}{9}, \frac{1}{25}, \frac{1}{49}, \frac{1}{81}$

$2(1)\ \ 2(2)\ \ 2(3)\ \ 2(4)$

$C \Rightarrow \left\{x : x \Rightarrow \frac{1}{(2n+1)^2};\ n \in N,\ n \leq 4\right\}$

Ex. The no. of Elements in the Set $A \Rightarrow \{(a,b) : 2a^2 + 3b^2 \Rightarrow 35,\ a, b \in Z\}$, where Z in the set of all Intgeas is

→ Given,

1) $A \Rightarrow \{(a,b) : 2a^2 + 3b^2 \Rightarrow 35,\ a, b \in Z\}$

2) $a, b \in Z$

$Z \Rightarrow (-\infty, 0, \infty)$

$\frac{1}{81}$ }

Evaluating an Expression

1) $2a^2+3b^2 \Rightarrow 35$

When $a \Rightarrow 0$

$2(0)^2+3b^2 \Rightarrow 35$

$b^2 \Rightarrow \frac{35}{3}$

$b \Rightarrow \sqrt{\frac{35}{3}}$

{x}

(2) $2a^2+3b^2 \Rightarrow 35$

$a \Rightarrow 1$

$2(1)+3b^2 \Rightarrow 35$

$3b^2 \Rightarrow 33$

$b^2 \Rightarrow 11$

$b \Rightarrow \sqrt{11}$

{x}

3) $2a^2+3b^2 \Rightarrow 35$

$(a \Rightarrow 2) 2(2)^2+3b^2 \Rightarrow 35$

$3b^2 \Rightarrow 35-8$

$b^2 \Rightarrow \frac{27}{3}$

$b \Rightarrow \sqrt{9} \Rightarrow \pm 3$

$a \Rightarrow \pm 2$

$b \Rightarrow \pm 3$

(✓)

4) $2(a)^2+3b^2 \Rightarrow 35$

$a \Rightarrow 3$

$2(3)^2+3b^2 \Rightarrow 35$

$18+3b^2 \Rightarrow 35$

$3b^2 \Rightarrow 35-18$

$b^2 \Rightarrow \frac{17}{3}$

(x)

5) $2a^2+3b^2 \Rightarrow 35$

$a \Rightarrow 4$

$2(4)^2+3b^2 \Rightarrow 35$

$3b^2 \Rightarrow 35-32$

$3b^2 \Rightarrow 3$

$b \Rightarrow \sqrt{1}$

$b \Rightarrow \pm 1$

$a \Rightarrow \pm 4$

$(a,b) \Rightarrow \{(-2,-3)\ (-2,3),(2,-3),(2,3),$
$(-1,-4)\ (-1,4),(1,-4),(1,4)\}$

Total no. of Elements $\Rightarrow 8$

Types of Set

1) Null Set (or) Empty Set

A Set having no Element in it. It is denoted by ϕ (or) $\{\}$.

Ex:

$A \Rightarrow \{x : 0 < x < 1 ; x \in \underbrace{I}_{\text{Integer}}\}$

2) Singelton Set

→ Set Consisting of a Single Element is Called a Singelton Set.

Ex:

$A \Rightarrow \{x : 0 < x < 2 ; x \in Z\}$

$A \Rightarrow \{1\}$

3) Finite Set

A Set which has Countable no. of Elements Called a finite Set.

Ex:

Set of prime numbers Len than 10.

By φ(or)

* → To Count How many number Exist We Call ~~the~~ it.

→ Order of a finite Set:
The number of Elements in a finite Set A Called the order of this Set and denoted by O(A) (or) n(A). It also Called Cardinal no. of Set.

A ⇒ {a, b, c, d}
n(A) ⇒ 4

4) Infinite Set

A Set Which has an infinite no. of Element in Called an Infinite Set.

N: {1, 2, 3, ...}
W: {0, 1, 2, ...}
Z: {-∞, 0, ∞}

5) Equal Set

Two Set A and B are Said to be Equal if every Element of A in member of B, and Every Element of B in a member of A.

∴ A ⇒ {1, 2, 3}
∴ B ⇒ {3, 2, 1}

{A ⇔ B}

6. Equivalent Sets

Two finite Set A and B are Equivalent if their no. of elements are same i.e $n(A) = n(B)$

e.g $A \Rightarrow \{1, 3, 5, 7\}$, $B \Rightarrow \{a, b, c, d\}$
$n(A) \Rightarrow 4$ and $n(B) \Rightarrow 4$

"A" and "B" are Equivalent Sets.

7) Universal Set

A Set Consisting of all possible Elements which occur in the discussion is called a Universal Set and is denoted by "U".

Ex:

If $A \Rightarrow \{1, 2, 3\}$, $B \Rightarrow \{2, 4, 5, 6\}$

$C \Rightarrow \{1, 3, 5, 7\}$ then

$\therefore U \Rightarrow \{1, 2, 3, 5, 6, 7\}$ Can be taken as the Universal Set.

Subset and Superset

Let A and B be two Set. if Even
of A is ~~Equivalent~~ an Element B then
called a Subset of B and B is Calle
of A. We write it as $A \subseteq B$

E.g:
$A \Rightarrow \{1, 2, 3, 4\}$ and
$B \Rightarrow \{1, 2, 3, 4, 5, 6\}$

$A \subseteq B$

If A is not a Subset of B then We
$A \not\subseteq B$.

Proper Subset

If A is a Subset of B but $A \neq B$
a proper Subset of B and We
. Set A is not proper Subset of A
improper Subset of A.

Ex. Write the proper Subset of $A \Rightarrow \{$

→ $A \Rightarrow \{1, 2, 3\}$

Subset ⇒

$\{\{\}, \{1\}, \{2\}, \{3,\}, \{1, 2\}, \{2, 3\}\ \{3, 1\}, \{1, 2, 3\}$

Proper Subset

$\{\{\}, \{1\}, \{2\}, \{3\}, \{1, 2\}\ \{2, 3\}, \{3, 1\}\} \Rightarrow B$

$\subseteq$

B. is a proper Subset.

Ex Let $A \Rightarrow \{1, 2, \{3, 4\}, 5\}$. Which of the following Statement is Correct.

→ $A \Rightarrow \{1, 2, \{3,4\}, 5\}$

~~$\{3,4\}$~~

$\{3,4\} \in A$

Important Results

1) Every Set is a Subset of itself.
2) Empt Set ; ϕ ; is a Subset of Every set.
3) $A \subseteq B$ and $B \subseteq A$ then $A \Leftrightarrow B$
4) ϕ is a proper set of every set except itself.

✳

5) The total no. of Subsets of a finite Set Containing "n" elements is 2^n.
6) The total no. of Proper Subsets of a finite Set Containing "n" elements in "$2^n - 1$"!

of the

Ex. If $A \Rightarrow \{x : x \Rightarrow 4n+1, 2 \leq n \leq 5\}$, then no. of Subsets of A is.

Given,

→ $A \Rightarrow \{x : x \Rightarrow 4n+1, 2 \leq n \leq 5\}$

1) $x : x \Rightarrow 4n+1, 2 \leq n \leq 5$

$n \Rightarrow 2, 3, 4, 5$

$x \Rightarrow 4n+1$

2) $n \Rightarrow 2$ $x \Rightarrow 4(2)+1$ $x \Rightarrow 9$

$n \Rightarrow 3$ $x \Rightarrow 4(3)+1$ $x \Rightarrow 13$

$n \Rightarrow 4$ $x \Rightarrow 4(4)+1$ $x \Rightarrow 17$

$n \Rightarrow 5$ $x \Rightarrow 4(5)+1$ $x \Rightarrow 21$

$A \Rightarrow \{9, 13, 17, 21\}$

no. of Subset $\Rightarrow 2^n \Rightarrow 2^4 \Rightarrow 16$

$n \Rightarrow$ no. of Elements $\Rightarrow 4$

Ex Let $A \Rightarrow \{a, b, c, d\}$ and $B \Rightarrow \{a, b, c\}$. The the no. of Sets X Contained in A and not Conte -ed in B is

→ Given,

$A \Rightarrow \{a, b, c, d\}$

$B \Rightarrow \{a, b, c\}$

X in a Set that Contains the Element that we get after the Subset of A and B. Where X in Containing the elements that are Subset of A but not B.

1) No. of Subset of $A \Rightarrow 2^n \Rightarrow n \Rightarrow 4 \Rightarrow 2^4 \Rightarrow 16$

no. of Subset of $B \Rightarrow 2^n \{n \Rightarrow 3\} \Rightarrow 2^3 \Rightarrow 8$

$X \Rightarrow 16 - 8 \Rightarrow 8$

Ex Two finite Set have m and n elements. The number of Subset of the first Set 112 more than that of the Second set. The values of m and n are respectively

→ Given,
m and n are Elements
Subset of m elements 112
$m > n$

1)→ A ⇒ no. of elements ⇒ m =
B ⇒ no. of elements ⇒ n

2^m ⇒ no. of Subset
2^n ⇒ no. of Subset

$2^m \Rightarrow 112$
$2^m - 112 \Rightarrow 2^n$
$2^m - 2^n \Rightarrow 112$
$2^n(2^{m-n} - 1) \Rightarrow 112$

→ LCM

2	112
2	56
2	28
2	14
7	7
	1

2→ $2^n(2^{m-n} - 1) \Rightarrow 2^4(7)$
$2^n \Rightarrow 2^4$
$n \Rightarrow 4$

umbe
han
dn

8) $2^{m-n} - 1 \Rightarrow 7$

$n \Rightarrow 4$

$2^{m-4} \Rightarrow 8$

$\frac{2^m}{2^4} \Rightarrow 2^3$

$2^m \Rightarrow 2^{3+4}$ $\{2^m \Rightarrow 2^3 \times 2^4 \{a^m \times a^n \Rightarrow a^{m+n}\}\}$

$2^m \Rightarrow 2^7$

$m \Rightarrow 7$

Power Set

- Let A be any Set. The "Set of all Subsets" of A in called power Set of A and is denoted by P(A).

Ex) find Power Set of set A $\Rightarrow \{1,2\}$

→ $A \Rightarrow \{1,2\}$

Subset of $A \Rightarrow \{1,2\}, \{\}, \{1\}, \{2\}, \$$

Power Set of $A \Rightarrow \{\{\}, \{1\}, \{2\}, \{1,2\}\}$

Ex. The no. of Subsets of the Power Set of Set $A \Rightarrow \{a, b, c\}$ is

$\Rightarrow$

$A \Rightarrow \{a, b, c\}$

no. of element $\Rightarrow 3$

Subset $\Rightarrow 2^3 \Rightarrow 8$

$P(A) \Rightarrow$ 8 elements in $P(A)$

Subset of $P(A) \Rightarrow 2^8 \Rightarrow 256$

Ex If A is a null set and B ⇒ P(P(P(A)))
Where P(A) denotes power set of A, then
which of the following is/are Correct.

→ Given,
A = {{}} (or) ϕ
B ⇒ P(P(P(A)))

Subset A ⇒ {}, ϕ
B = P(A) ⇒ {{}, {ϕ}} (or) {ϕ, {ϕ}}
B Subset ⇒ {}, {ϕ},

Given,
A ⇒ ϕ
Subset A ⇒ {}, ϕ

1) P(A) ⇒ {{}, ϕ}
P(A) ⇒ B

2) B of Subset ⇒ {}, ϕ, {ϕ}, {ϕ, {ϕ}}
C = P(B) ⇒ {{}, ϕ, {ϕ}, {ϕ, {ϕ}}}

3) C ⇒ {{}, ϕ, {ϕ}, {ϕ, {ϕ}}}
P(C) ⇒ {}, ϕ, {ϕ}, {{ϕ}}, {ϕ, {ϕ}},
{ϕ, {ϕ}}, {ϕ, {{ϕ}}}, {ϕ, {ϕ, {ϕ}}},
---16--

Venn Diagram

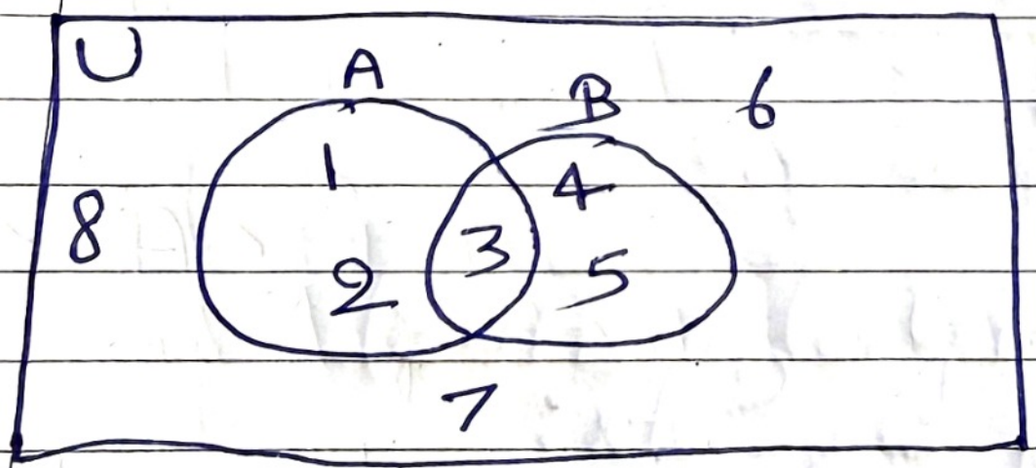
→ Most of the relationships b/w sets can be represented by means of diagrams which are known as "Venn Diagrams"

Ex.

If

A ⇒ {1, 2, 3},
B ⇒ {3, 4, 5},
U ⇒ {1, 2, 3, 4, 5, 6, 7, 8}

then the Venn Diagram is



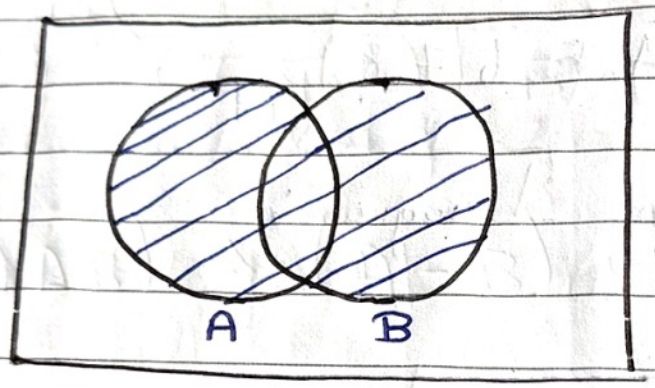
Operation of Sets

1) Union of two sets:

$A \cup B \Rightarrow \{x : x \in A \text{ (or) } x \in B\}$

* E.g : $A \Rightarrow \{1,2,3\}$, $B \Rightarrow \{2,3,4\}$

then $\Rightarrow A \cup B \Rightarrow \{1,2,3,4\}$



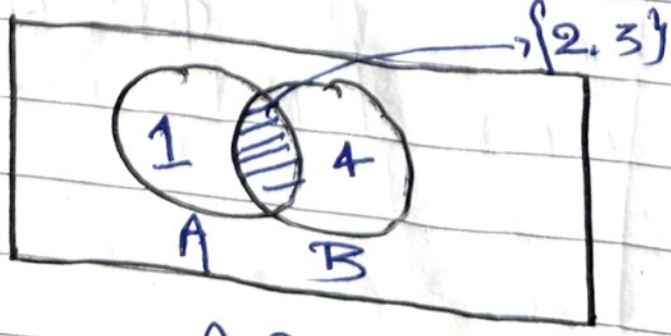
$A \cup B$

2) Intersection of two Sets:

$A \cap B \Rightarrow \{x : x \in A \text{ and } x \in B\}$

E.g : $A \Rightarrow \{1,2,3\}$, $B \Rightarrow \{2,3,4\}$

$A \cap B \Rightarrow \{2,3\}$

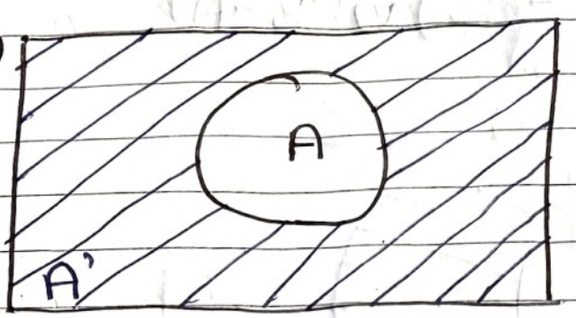


$A \cap B$

3) Complement of Set: $A' \Rightarrow \{x : x \notin A \text{ but } x \in U\}$

e.g: $U \Rightarrow \{1, 2, \ldots, 10\}$, $A \Rightarrow \{1, 2, 3, 4, 5\}$ then

$A' \Rightarrow \{6, 7, 8, 9, 10\}$



4) Difference of two Sets:

$A - B \Rightarrow \{x : x \in A \text{ and } x \notin B\}$

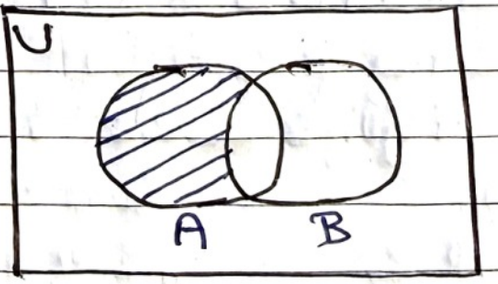
It is also Written as $A \cap B'$ {elements that are common b/w A and B}

e.g: $A \Rightarrow \{1, 2, 3\}$, $B \Rightarrow \{2, 3, 4\}$

$A - B \Rightarrow \{1\}$

Similarly

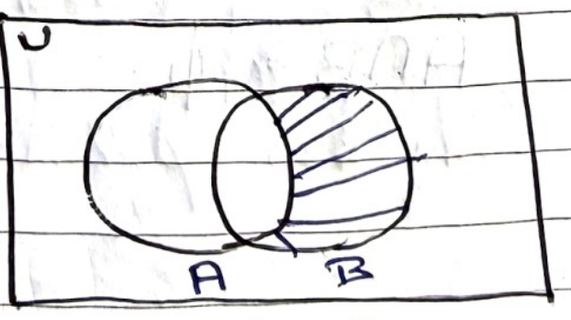
$B - A \Rightarrow B \cap A'$



A - B

* Can also be Written as:

$A - B \Rightarrow A - (A \cap B)$



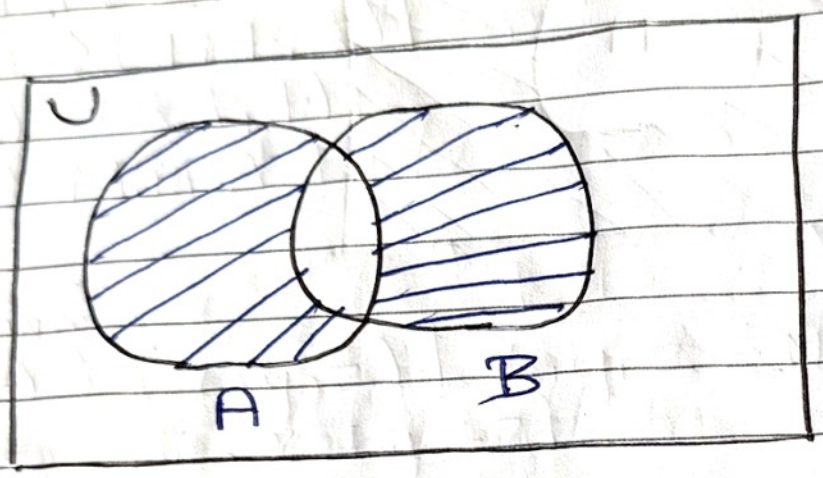
B - A

5) Symmetric difference of two Sets:

→ It is denoted by 'A Δ B' and

$A \Delta B \Rightarrow (A - B) \cup (B - A)$

Ex



$(A \Delta B) \Rightarrow (A - B) \cup (B - A)$

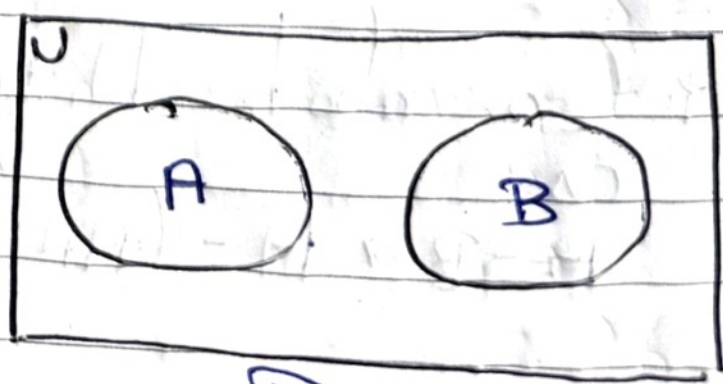
It also be Written as

→ $A \Delta B \Rightarrow (A \cup B) - (A \cap B)$

) Disjoint Sets: If $A \cap B \Rightarrow \phi$. then A and B are disjoint.

e.g: $A \Rightarrow \{1,2,3,4\}$. $B \Rightarrow \{6,7,8,9\}$ Then

$A \cap B \Rightarrow \phi$



Disjoint

⇒ What is a prime factor?

A "prime factor" is a number that is also an "prime number"

Ex. Take the number 12

The factor of 12 are

1, 2, 3, 4, 6, 12

⇒ The prime numbers among these factor are 2 and 3

So, the prime factor of 12 are 2 and 3

- Prime factor of 12 → 2, 2, 3
- Unique prime factor of 12 → 2, 3
- Prime factorization of 12 ⇒ $2^2 \times 3$

Ex Let A ⇒ {x : x is a prime factor of 240}
B ⇒ {x : x is the sum of any two prime factor of 240}

→ 240

2	240
2	120
2	60
2	30
3	15
5	5
	1

240 ⇒ 2, 2, 2, 2, 3, 5

A ⇒ {2, 3, 5}

B ⇒ {5, 7, 8}

